Rourou Chen

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RESEARCH INTERESTS

3D Scene Reconstruction & Digital Twins: editable, metric-consistent driving scenes reconstructed from real multi-sensor captures.
Controllable Generative Models: geometry-aware synthesis of long-tail objects and adverse weather as labeled training and test data.
Robust 3D Perception: evaluating and hardening autonomous-driving perception against out-of-distribution corner cases.

EDUCATION

Sun Yat-sen University **Sep. 2024 – Jun. 2027 (Expected)**

- M.Eng. in Electronic Information, School of Intelligent Systems Engineering, Shenzhen
- Advisor: Prof. Xiaojun Tan | Weighted Average: 86.8/100 | GPA: 3.50/5.0
- Coursework: *Multi-Agent Systems* (95.0), *Image Processing & Machine Vision* (92.5), *Computer Vision* (91.0), *Advanced Algorithm Design & Analysis* (89.9), *Intelligent Vehicles & ADAS* (85.6)

Sun Yat-sen University **Sep. 2020 – Jun. 2024**

- B.Eng. in Intelligence Science and Technology, School of Intelligent Systems Engineering
- GPA: 3.50/5.0

MANUSCRIPT UNDER REVIEW

[Full author list in submitted order]. RECAST: Reference-Guided Calibrated Augmentation of Object States for Monocular 3D Detection. *IEEE Robotics and Automation Letters (RA-L)*, under review.

RESEARCH EXPERIENCE

RECAST: Reference-Guided Generation for Long-Tail Monocular 3D Detection **Feb. 2026 – Present**
Lead Researcher | *Manuscript under review at IEEE RA-L*

- Targeted monocular 3D detection failures on long-tail states (rare orientations, distant vehicles) with a reference-guided diffusion pipeline driven by 2D exemplars and target 3D boxes, without CAD assets.
- Designed structured cuboid-face conditioning for geometric control and a detector-in-the-loop recalibration stage that re-fits each synthetic label to the generated object.
- On nuScenes (FCOS3D), **22.9K** synthetic instances raised mAP by **1.2** and Rare-5 AP by **1.1** points over real-replay baselines; synthetic failure modes tracked real ones, enabling targeted stress testing.

GeoWeatherEdit: Label-Preserving Adverse-Weather Scenario Generation **May 2026 – Present**
Graduate Research Project

- Developed a training-free pipeline that synthesizes snow, rain, and fog while preserving instance geometry and 3D labels, decoupling constrained 3D instance compositing (SAM3D assets, road, ground-contact, occupancy, and visibility checks) from one-step diffusion weather transfer.
- Outperformed WeatherEdit, DriveGEN, and ChordEdit on background-structure preservation (Edge F1 **0.531**) with top CLIP-direction alignment (**0.162**); downstream 3D detection benchmarking underway.

Sensor-to-Twin: Real-World Gaussian Digital Twins with Geometry-Grounded Editing **Sep. 2024 – Present**
Master's Thesis | *Shenzhen Key Research Project*

- Built a pipeline that turns self-collected campus captures into Gaussian digital twins across three sensor setups: images only (3DGS), vehicle cameras + LiDAR (Street Gaussians), and ground + UAV imagery (HorizonGS).
- Traced floating Gaussians and dynamic residue to a $\sim$**90 ms** RGB–LiDAR desynchronization and noisy initialization; suppressed them via temporally windowed reconstruction and point-cloud filtering.
- Recovered metric scale through lane geometry and inverse perspective mapping, chaining COLMAP–map–Blender transforms to enforce road, collision, and visibility constraints on object placement.

- Re-engineered an open-source LLM multi-agent editor into a backend-agnostic engine with a **three-layer simulation stack**: scene (free camera), vehicles (insertion, removal, trajectories), and weather.

Tool-Augmented LLM Agents for Cooperative Highway Merging **Jun. 2025**

Independent Project, Multi-Agent Systems Course

- Extended HighwayEnv into an on-ramp merging scenario with mixed connected (CAV) and human-driven (HDV) traffic at 25/50/75% CAV penetration, and built an LLM decision pipeline equipped with four tools: legal-action, reachable-lane, and neighbor-state queries plus trajectory-conflict checks.
- Designed priority-ordered sequential decisions with CAV intent sharing, pairing high-level LLM decisions with rule- and trajectory-based safety verification; in an 8-vehicle scenario, achieved **100%** merge success at every penetration rate with **zero** potential conflicts (5 m threshold), against rule-based, virtual-platoon, and game-theoretic baselines.

SELECTED UNDERGRADUATE RESEARCH

Video Anomaly Detection for Exam-Room Surveillance **Dec. 2023 – Jun. 2024**

Undergraduate Thesis

- Built a multi-person tracking and action-segmentation pipeline (YOLOv5, DeepSORT) over exam-room video and applied Vision Transformers to classify actions and flag anomalous examinee behavior.

Flexible Tactile Sensing for Surface Roughness Perception **Dec. 2021 – Nov. 2022**

Undergraduate Innovation Project

- Developed a flexible magnetic tactile sensor and XY-stage platform, acquiring signals with Arduino and Python to reconstruct 3D surface profiles; second-listed contributor to a related vision–tactile patent application [application no. to add].

HONORS AND AWARDS

- **2024, 2026** Graduate Academic Scholarship (Second Class), Sun Yat-sen University
- **2023** National Encouragement Scholarship
- **2023** Outstanding Student Scholarship (Third Class), Sun Yat-sen University
- **2022** Provincial Second Prize (Guangdong), China Undergraduate Mathematical Contest in Modeling
- **2022** Provincial Third Prize, Guangdong / Guangdong–Hong Kong–Macao College Students' Engineering Practice and Innovation Competition

ATHLETICS, LEADERSHIP AND SERVICE

Varsity Athlete, Sun Yat-sen University Women's Volleyball Team **2021 – Present**

- **Two-time Champion**, Guangdong Provincial University Games, Women's Volleyball (2021, 2025)
- **Champion**, Guangdong Provincial Volleyball League, Division A (2024); **Runner-up**, National University Volleyball League, Sunshine Division (2025)

Leadership and Service

- Deputy Head, Refereeing Division, Sun Yat-sen University Shenzhen Volleyball Association
- Volunteer, Chinese Women's Volleyball Super League (Shenzhen); volunteer, elderly-care home visits

SKILLS

3D Vision & Reconstruction: 3D Gaussian Splatting (3DGS, Street Gaussians, HorizonGS), COLMAP, Open3D, LiDAR–camera calibration, IPM and multi-frame coordinate transforms

Generative Models: diffusion-based editing (Diffusers, one-step ChordEdit), SAM / SAM3D, Grounded-SAM

Perception: PyTorch, MMDetection3D (FCOS3D), YOLOv5, DeepSORT, OpenCV

Simulation & Robotics: LLM multi-agent pipelines, Blender, CARLA, ROS, Arduino

Programming: Python, C++, MATLAB, Linux

Languages: Mandarin (native); English (CET-6: 521)